A Controlled Evaluation of Curvature-Inspired Scalar Feature Scaling for Domain Generalization

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Abstract

Domain-specific curvature is an appealing idea for domain generalization (DG), but *curvature learning* can denote substantially different computations. This paper evaluates one deliberately narrow implementation: the code calls it HyperDG, but it rescales Euclidean encoder features during training by learned positive scalars and predicts with an unscaled Euclidean feature and an ordinary linear classifier. It does not implement a Lorentz or Poincaré representation, geodesic prediction, or a Sharpness-Aware Minimization inner loop. We audit the executed objective and complete a six-condition, four-dataset (PACS, VLCS, OfficeHome, TerraIncognita), four-target, four-seed evaluation (384 runs), with checkpoints selected using only the three source-validation environments and target-level effects, not 16 seed-by-target cells, as the per-dataset inferential unit. The full implementation's equal-dataset macro estimate is 1.61 percentage points below native DomainBed ERM; all four dataset estimates favor ERM, but the primary four-dataset exact sign-flip test gives $p = .125$ (Holm-adjusted $p = .250$), so we claim no adjusted evidence of a difference. The cleaner domain-specificity contrast, separate learned source scales without their dispersion penalty versus one shared learned source scale, is $-0.13\,pp$ and passes a revision-stage $\pm 1\,pp$ equivalence test (TOST $p = .003$) for the source-selected suite macro, though not for every dataset, target, or the final-step endpoint. The study thus supplies a bounded negative result for this scalar training package; it does not test or refute genuine hyperbolic representation learning.

1 Introduction

Domain generalization (DG) trains on several source environments and evaluates on an unseen target environment (Wang et al., 2023; Gulrajani & Lopez-Paz, 2021). The setting invites geometric inductive biases: hyperbolic spaces can represent hierarchies and multiple scales compactly, and modern hyperbolic networks can operate directly on a curved manifold (Ganea et al., 2018; Fan et al., 2025). A much weaker intervention is to multiply an otherwise Euclidean feature vector by a learned scalar. Those interventions should not be evaluated or described as if they were interchangeable.

This distinction is central here. *In this paper,* HyperDG denotes the exact scalar-scaling formulation specified in Section 3; it is not a reference to a separate external paper and it is not shorthand for hyperbolic DG in general. The implementation maps a Euclidean feature $h_\theta(x)$ to $h_\theta(x)/\sqrt{c_i}$ during its training loss, where $c_i > 0$ is indexed by a source-loader slot. The model predicts with the unscaled Euclidean feature and a

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linear classifier. Calling c_i a curvature magnitude reflects the implementation's motivation, not a claim that the representation has learned sectional curvature.

We ask two bounded questions:

1. Does the complete HyperDG training package improve held-out-domain accuracy over native DomainBed ERM under a matched ResNet-50 protocol?
2. Within the same HyperDG update route, do three separately learned source scales provide a practically meaningful macro advantage over one shared learned source scale?

The first comparison is a whole-package utility test, because native ERM and HyperDG differ in update and BatchNorm routes. The second isolates the per-source parameterization more cleanly by comparing NoVariance with SharedLearnable; neither condition has an operative between-scale dispersion penalty.

This revision contributes:

- an implementation-level specification that separates the intended geometric interpretation from the executed computation, including an inactive local-gradient term, an algebraically scale-cancelling proxy, and the train–inference scaling mismatch;
- a completed six-condition, four-dataset, four-target, four-seed evaluation (384/384 runs), including native ERM and a fully replicated shared-learnable condition;
- source-only model selection and target-clustered uncertainty, with every named target and run seed reported separately; and
- a narrow negative conclusion: under this protocol, the full package shows no advantage over ERM, and separate source scales are macro-equivalent to one shared scale within the stated revision-stage margin.

2 Related Work and Scope Boundary

Domain generalization and ERM. ERM minimizes average source risk (Vapnik, 1998). DomainBed showed that controlled model selection and protocol matching can erase many claimed DG gains (Gulrajani & Lopez-Paz, 2021). Proposed DG objectives include invariance, risk extrapolation, and distribution alignment (Arjovsky et al., 2019; Krueger et al., 2021; Sun & Saenko, 2016). We therefore compare against unmodified DomainBed ERM rather than importing literature numbers, and we keep target-domain data out of the checkpoint score.

Genuine hyperbolic representations. Hyperbolic neural networks implement manifold-valued operations and hyperbolic classifiers (Ganea et al., 2018). MERU learns image–text representations on a Lorentz hyperboloid (Desai et al., 2023). Recent work learns curvature within hyperbolic networks (Fan et al., 2025), and DG-specific methods use hyperbolic state space hallucination or cross-branch consistency (Bi et al., 2025; Di Salvo et al., 2026). These are materially stronger geometric parameterizations than the scalar rescaling audited here. They are also different algorithms, tasks, architectures, and selection protocols, so their published numbers are not valid matched controls for our experiment. A controlled comparison would require implementing a manifold-valued encoder or head in the same 384-cell protocol; the present evidence does not contain such an experiment.

3 Method Under Test

3.1 Executed feature scaling

For source-loader slot i, the encoder produces $h_i = h_\theta(x_i) \in \mathbb{R}^{2048}$. A free scalar a_i is transformed to

$$c_i = \mathrm{softplus}(a_i) + 10^{-4}, \qquad z_i = \frac{h_i}{\sqrt{c_i}}. \tag{1}$$

Table 1: The claim boundary. The last row is the only mechanism tested in this paper. Hyperbolic in the other rows denotes manifold-valued operations or representations, not merely a scalar applied to Euclidean features.

Family	Representation and prediction	Relation to this study
Hyperbolic neural networks (Ganea et al., 2018)	Poincaré/Möbius operations and a hyperbolic classifier	Genuine manifold baseline; not implemented here
MERU and learned-curvature HNNs (Desai et al., 2023; Fan et al., 2025)	Lorentz or hyperbolic embeddings with explicit manifold geometry	Stronger geometric parameterization; different protocol
Hyperbolic DG (Bi et al., 2025; Di Salvo et al., 2026)	Hyperbolic representations embedded in DG-specific objectives	Relevant recent direction; not a scalar-only ablation
HyperDG (this audit)	$h_\theta(x)/\sqrt{c_i}$ in training; Euclidean linear prediction on $h_\theta(x)$	Curvature-inspired scalar feature scaling only

All a_i and a_g start at zero, hence $c_i = c_g = \log 2 + 10^{-4} = 0.693247$ at initialization, giving a feature multiplier $1/\sqrt{c} = 1.201036$. In MatchedControl and Frozen the scale optimizer's learning rate is set to zero, so a_i and a_g never move from zero and $c_i = c_g = 0.693247$ is the constant value used for every update of the entire run, not merely the initial value. In FullPerDomain, NoVariance, and SharedLearnable, a_i (and, for SharedLearnable, the one shared a) and a_g are learned from this same starting point. A separate learned c_g defines $z_{g,i} = h_i/\sqrt{c_g}$. In a leave-one-domain-out run there are three source loaders. Consequently, per-domain in the code means three source-loader slots, whose semantic domain mapping changes with the held-out target; there is no parameter for the unseen target.

The local supervised loss is

$$\mathcal{L}_{\text{sup}} = \frac{1}{3}\sum_i \text{CE}(W z_i + b, y_i). \tag{2}$$

The code also forms domain-mean alignment

$$\mathcal{L}_{\text{align}} = \frac{1}{3}\sum_i \left\| \bar{z}_i - \sum_j w_j \bar{z}_j \right\|_2^2, \tag{3}$$

where $\bar{z}_i = \frac{1}{|B_i|}\sum_{k\in B_i} z_{i,k}$ is the mean of the scaled features over the $|B_i| = 32$ examples in source minibatch i, and $w_j = B_j / \sum_k B_k$ weights each source by its minibatch size (a 10^{-8} denominator stabilizer is otherwise negligible). Every source batch has 32 examples here, so $w_j = 1/3$ for all j; the alignment term reduces to pulling each domain mean toward the unweighted mean of the three domain means. Pairwise scale dispersion is

$$\mathcal{L}_{\text{var}} = \frac{1}{9}\sum_{i,j}(c_i - c_j)^2, \tag{4}$$

and a global disagreement term

$$\mathcal{L}_{\text{global}} = \frac{1}{3}\sum_i \left|\text{CE}(W z_{g,i} + b, y_i) - \text{stopgrad}[\text{CE}(W z_i + b, y_i)]\right|. \tag{5}$$

3.2 Intended versus executed auxiliary terms

The executed code matters for interpretation. First, the quantity named the norm/sharpness proxy is

$$\mathcal{L}_{\text{norm}} = \frac{1}{3}\sum_i c_i \ \text{Var}\big(\|h_i/\sqrt{c_i}\|_2\big) = \frac{1}{3}\sum_i \text{Var}(\|h_i\|_2), \tag{6}$$

away from the inactive numerical clamp, where $\text{Var}(\cdot)$ is the population variance (divisor $|B_i| = 32$, PyTorch `unbiased=False`) of the 32 per-example scaled-feature norms $\|h_{i,k}/\sqrt{c_i}\|_2$, $k \in B_i$, computed separately

within each source minibatch i before the three domain values are averaged. Because $\sqrt{c_i}$ is constant across the examples in minibatch i, it factors out of the variance and squares back in against the multiplying c_i, which is why the expression collapses to $\mathrm{Var}(\|h_i\|_2)$: the term regularizes raw feature-norm dispersion but its value does not depend on c_i, so it is not curvature-sensitive and it is not SAM (Foret et al., 2021).

Second, the intended local term differentiates each supervised loss with respect to a_i. The implementation requests that gradient from a newly indexed view that is not the view used to construct the forward scale. Automatic differentiation therefore returns `None`, which is converted to zero:

$$\mathcal{L}_{\text{local}}^{\text{executed}} = 0. \tag{7}$$

Every primary-run log records this term as zero. This is an implementation defect, not evidence that a correctly implemented curvature sensitivity is zero.

The executed total loss is

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + \alpha\mathcal{L}_{\text{var}} + \beta\mathcal{L}_{\text{align}} + \gamma\mathcal{L}_{\text{norm}} + \delta_\ell\mathcal{L}_{\text{local}} + \delta_g\mathcal{L}_{\text{global}}. \tag{8}$$

The full coefficients are $(\alpha, \beta, \gamma, \delta_\ell, \delta_g) = (0.2, 0.5, 0.05, 0.1, 0.1)$, but the local contribution is identically zero. Finally, `predict()` applies $Wh_\theta(x) + b$, not $Wz + b$. Thus scales act only through training; they are not used for target prediction. With a linear head, a common global scale is also partly absorbable into W.

3.3 Six controlled conditions

The six conditions in Table 2 play two different inferential roles, and only two contrasts among them are primary claims of this paper. FullPerDomain−StandardERM and NoVariance−SharedLearnable are the two primary suite contrasts defined in Section 4 and carried through Holm correction in Section 5. MatchedControl and Frozen are secondary, descriptive diagnostics: no hypothesis test in this paper treats a contrast involving either condition as a primary claim. MatchedControl runs the HyperDG update route with a constant, non-learned feature rescaling and every auxiliary coefficient set to zero, so it shows what the route alone costs or gains relative to StandardERM before any HyperDG-specific loss term is active. Frozen is the fixed-scale counterpart of NoVariance: it keeps the same non-dispersion auxiliary losses active but never lets the scales move, so comparing it with NoVariance descriptively probes learned against fixed scaling. Frozen is not a substitute for SharedLearnable, because it changes learnability rather than per-source specificity; NoVariance−SharedLearnable is the condition pair that isolates the domain-specificity question.

Table 2: Completed conditions. All five non-ERM conditions use the same HyperDG update route. Only StandardERM, FullPerDomain, NoVariance, and SharedLearnable enter the paper's two primary contrasts; MatchedControl and Frozen are descriptive diagnostics (see text).

Condition	Executed treatment
StandardERM	Native DomainBed ERM; no scales or auxiliary losses.
MatchedControl	HyperDG route; three fixed local scales and one fixed global scale; all five coefficients zero.
FullPerDomain	Three learned local scales and one learned global scale; full coefficients in Equation 8.
Frozen	All scales fixed; retains alignment, norm, local (inactive), and global terms; $\alpha = 0$.
NoVariance	Three learned local scales; same remaining objective as FullPerDomain but $\alpha = 0$.
SharedLearnable	One learned local scale shared by all three source loaders plus the separate learned global scale. Scale dispersion is structurally zero.

4 Experimental Design

Benchmarks and why they are informative. We use PACS (Li et al., 2017), VLCS (Fang et al., 2013), OfficeHome (Venkateswara et al., 2017), and TerraIncognita (Beery et al., 2018). Their four environments

emphasize different shift types: depiction/style (PACS), acquisition and dataset source (VLCS), depiction plus object context (OfficeHome), and camera location/background (TerraIncognita). This diversity is useful for asking whether one scalar specializes consistently across source environments. None of these datasets provides ground-truth manifold curvature, hierarchy, or a known geometric scale, so they cannot by themselves reveal that a representation's true curvature differs by domain.

Protocol. Each environment is held out once; the other three are sources. Within every environment, 80/20 splits are deterministic functions of the run seed and environment index. Seeds 0–3 vary data splits, initialization, minibatch order, and augmentation. All methods use `timm resnet50.ram_in1k` ImageNet weights, a 2,048-dimensional feature, a biased linear classifier, source batch size 32, Adam with network learning rate 5×10^{-5}, zero weight decay, data augmentation, 5,000 updates, and evaluation every 200 updates plus update 4999. BatchNorm is trainable. Learned scales have a separate Adam optimizer at 10^{-4} and the HyperDG route clips network and scale gradients to norm 5. By *native DomainBed ERM* we mean the unmodified upstream DomainBed `ERM` class (Gulrajani & Lopez-Paz, 2021), instantiated with the same data, splits, augmentation, timm ResNet-50 backbone, and source-only selector as every other condition: it concatenates the three source minibatches into one batch, computes a single cross-entropy loss, and takes one Adam step with one optimizer and no gradient clipping. It is the project's training and evaluation harness around an otherwise untouched DomainBed ERM update rule, not a claim that the surrounding software stack is an unmodified upstream installation.

This last distinction prevents a curvature-only causal reading of FullPerDomain−StandardERM. Per update, HyperDG calls the feature extractor six times: three source-minibatch forward passes build the training objective, and, after the optimizer step, three additional forward passes under `torch.no_grad()` recompute training accuracy for logging. `no_grad()` only suppresses gradient tracking; it does not put the model in evaluation mode. The model is never switched to `eval()` during HyperDG's update, so all six forward passes run in ordinary training-mode BatchNorm, updating the running mean, running variance, and batch counter exactly as any other training-mode forward pass. Native ERM instead concatenates the three source minibatches and calls the feature extractor once on all 96 examples, so its BatchNorm statistics update from one training-mode forward pass per step rather than HyperDG's six. The comparison therefore estimates the utility of the complete implementation, including this BatchNorm-update asymmetry, not an isolated curvature effect. The NoVariance−SharedLearnable comparison uses the same route on both sides, so this asymmetry cancels out of that contrast.

Source-only checkpoint selection. For dataset j, held-out target d, seed s, and checkpoint t, selection maximizes

$$q_{jds}(t) = \frac{1}{3} \sum_{r \neq d} \mathrm{Acc}^{\mathrm{out}}_{jdsr}(t). \tag{9}$$

Only source environments enter q; exact ties choose the earliest checkpoint. Held-out-target accuracy is read after t^* is fixed. The trainer did log target curves, so the precise claim is that target accuracy is not used by the frozen selection rule. Source `out` subsets inherit stochastic source augmentation and are evaluated once per checkpoint, a limitation shared by all conditions. Update-4999 results are a prespecified revision-stage sensitivity endpoint.

Estimand. The scientific claim is about the equal-weighted macro over exactly these four benchmarks (PACS, VLCS, OfficeHome, TerraIncognita) under this protocol, not about a population or superpopulation of all DG tasks that these four datasets are sampled from. We chose the suite for its heterogeneity of shift type (Section 4), not as a random draw from a broader task distribution, so every suite-level number below should be read as a fixed-suite macro, not a generalization to unseen benchmarks.

Estimands and uncertainty. Let Y_{mjds} be source-selected target accuracy (percentage points) for method m. For contrast $A - B$,

$$g_{jds} = Y_{Ajds} - Y_{Bjds}, \quad \bar{g}_{jd} = \frac{1}{4} \sum_{s} g_{jds}, \quad \Delta_j = \frac{1}{4} \sum_{d} \bar{g}_{jd}. \tag{10}$$

Table 3: Source-validation-selected target accuracy (%). Each cell averages four named targets after averaging four run seeds within target. Bold marks the highest point estimate in a dataset, not a significance claim.

Condition	PACS	VLCS	OfficeHome	TerraInc.
StandardERM	85.88	**77.19**	**70.21**	**47.13**
MatchedControl	84.22	75.38	70.05	44.14
FullPerDomain	84.87	76.42	68.52	44.17
Frozen	**86.14**	75.84	67.81	44.20
NoVariance	85.28	76.17	68.36	43.70
SharedLearnable	85.55	76.56	68.17	43.75

Seeds are averaged within target first, and targets are averaged within dataset second; the suite-level procedures below use only the four resulting dataset means Δ_j and do not separately repropagate seed- or target-level variance into the suite interval. That within-dataset variability is real and is reported descriptively (per-target effects, between-target SD, and within-target seed SD all appear in the supplement), but it enters the suite-level test only through the four dataset means it produced, not as an additional variance component.

The four named target effects $\bar{g}_{jd}$ are the per-dataset inferential units. We report an exact two-sided sign-flip test over all $2^4 = 16$ target assignments, a t_3 interval, and a four-cluster bootstrap sensitivity. The sign-flip test is exact conditional on treating the four target-level sign flips as exchangeable under the joint no-effect null (every one of the 16 sign patterns equally likely if $\bar{g}_{jd}$ has a distribution symmetric about zero under the null); it is a working symmetry assumption for a four-cluster benchmark suite, not a claim that the datasets were themselves randomly assigned to conditions, and it does not test whether dataset-level effects cancel under a suite-macro-only null. The smallest possible two-sided exact p is .125 with four clusters. At suite level, $\Delta_{\text{macro}} = \frac{1}{4}\sum_j \Delta_j$ and the four dataset effects are the clusters. The two primary suite contrasts, FullPerDomain−StandardERM and NoVariance−SharedLearnable, form one Holm family (Holm, 1979). Per-dataset tests are an eight-test sensitivity family.

We declare the exact sign-flip test with Holm correction as the primary inferential procedure for both suite-level difference claims; the t_3 interval and the bootstrap sensitivity are secondary, working-model summaries that assume approximate normality or resampling validity with only four clusters and do not override the primary test. Where the two disagree, as for FullPerDomain−StandardERM (Section 5), the Holm-adjusted exact result governs the paper's claim.

Practical equivalence uses TOST (Schuirmann, 1987) with a revision-stage smallest effect of practical interest (SESOI) of $\pm 1\,pp$, chosen after the submitted results were known but before the added ERM/shared grid completed. Margins of ± 0.5 and $\pm 2\,pp$ are sensitivities. TOST, like the t_3 interval, is a secondary working-model summary conditional on the same four-cluster suite macro; we do not interpret a nonsignificant difference-test result as equivalence, and equivalence conclusions are likewise scoped to this fixed four-dataset macro.

5 Results

5.1 Completed six-condition study

All 384 planned runs completed. Table 3 reports equal-target macro accuracies. Every named target and seed, along with between-target and within-target run-seed variability, appears in the supplement.

5.2 Primary utility and domain-specificity contrasts

Whole-package utility. FullPerDomain−StandardERM is -1.01, -0.77, -1.69, and $-2.96\,pp$ on PACS, VLCS, OfficeHome, and TerraIncognita. Thus all four point estimates favor ERM. The suite t_3 interval excludes zero, but with four dataset clusters the primary exact test is $p = .125$ and Holm-adjusted $p = .250$; per Section 4, this primary Holm-adjusted result, not the secondary t_3 interval, governs the paper's claim. We therefore do not claim adjusted evidence of a difference. The compatible range also does not support

Table 4: Primary suite-macro contrasts. Exact p-values sign-flip the four dataset effects; Holm adjusts the two primary tests. TOST uses the $\pm 1\,pp$ revision-stage SESOI.

Contrast	Effect [95% CI]	Exact / Holm p	TOST p	Interpretation
Full − ERM	-1.61 $[-3.17, -0.04]$	.125 / .250	.848	No adjusted evidence; equivalence not demonstrated
NoVar − Shared	-0.13 $[-0.53, 0.27]$	.375 / .375	.003	$\pm 1\,pp$ source-selected macro-equivalent

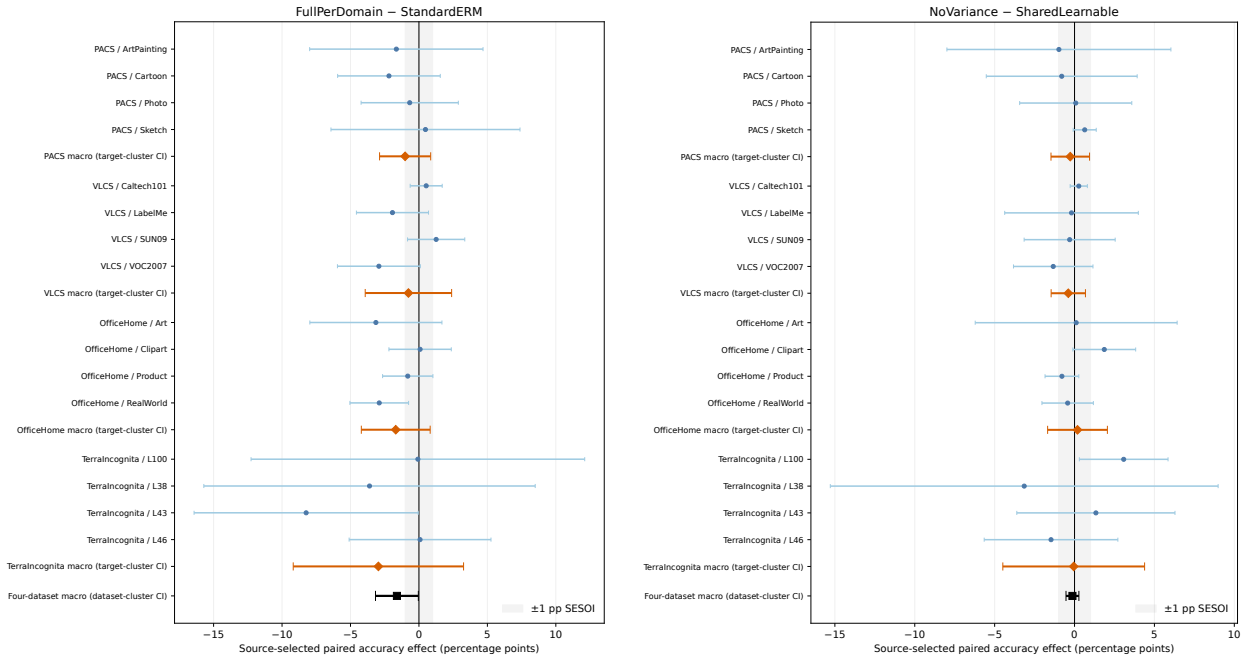


Figure 1: Paired effects for the two primary contrasts. Blue intervals are t_3 intervals across four paired run seeds within a named target; orange intervals cluster four target means within a dataset; black intervals cluster the four fixed dataset means. The visible target heterogeneity is why seed-by-target cells are not treated as 16 independent domain draws. These intervals describe this fixed, heterogeneous benchmark suite, not all DG tasks. Exact values and run-seed SDs are in the supplement.

equivalence of the packages. Relative to the HyperDG-path matched control, FullPerDomain changes the four dataset means by $+0.65$, $+1.04$, -1.53, and $+0.03\,pp$; this secondary pattern is mixed.

Are separate source scales necessary? The clean NoVariance−SharedLearnable effects are -0.27, -0.39, $+0.19$, and $-0.05\,pp$ across the same datasets. The suite effect is $-0.13\,pp$ with 90% CI $[-0.43, 0.17]$, wholly inside $\pm 1\,pp$ (TOST $p = .003$) and also inside the $\pm 0.5\,pp$ sensitivity margin (TOST $p = .031$). This supports a specific conclusion: for the source-selected, equal-dataset/equal-target macro estimand, three source-loader scales offer no practically meaningful advantage over one shared learned scale. We do not extend equivalence to any individual dataset or target, whose intervals are wider.

Checkpoint sensitivity. Source selection improves over update 4999 in 20 of 24 condition-by-dataset cells, by $0.97\,pp$ on average. At update 4999, NoVariance−SharedLearnable is $-0.32\,pp$ with 90% CI $[-1.13, 0.50]$ and TOST $p = .072$; equivalence is not established. We therefore restrict the equivalence conclusion to the declared source-selected endpoint.

Mechanistic audit. The local-gradient regularizer is zero in all primary logs, and Equation 6 shows that the norm proxy cannot identify curvature. The close performance of NoVariance and SharedLearnable is consequently consistent with a weakly identified, partly absorbable training-time scale. It should not be interpreted as evidence that a correctly implemented manifold-valued curvature parameter would behave similarly.

6 Discussion

The empirical answer is narrower than the original geometric motivation. First, the complete implementation provides no evidence of improvement over native ERM under the tested protocol: its four dataset effects are negative, while exact small-cluster inference prevents a strong adjusted difference claim. Second, within the common HyperDG route, separately learned source scales are macro-equivalent to one shared learned scale at the stated source-selected endpoint. This directly answers the parameterization question that Frozen could not answer: Frozen changes learnability and is not a substitute for a shared-learnable condition.

The dataset suite is useful for stress-testing scale specialization because it spans artistic style, dataset acquisition, object context, and camera-location shifts. The heterogeneity in Figure 1 also shows why target identity dominates some uncertainty summaries. But benchmark diversity is not geometric supervision. Without known hierarchies, distances, or target curvatures, a learned scalar cannot be validated as physical or sectional curvature merely because it differs between environments.

The comparison to recent hyperbolic methods is therefore conceptual rather than numerical. A genuine follow-up should keep the present splits, seeds, backbone, budget, and source-only selector while replacing the Euclidean head with a correctly implemented Poincaré or Lorentz representation and hyperbolic classifier. Fixed- versus learned-curvature versions would then separate manifold geometry from scalar flexibility. That experiment is scientifically worthwhile, but adding incomplete MERU runs or post-hoc Poincaré projections would not answer it; we exclude those diagnostics from the evidentiary claims.

7 Limitations and Broader Impact

Limitations. There are only four named target clusters per benchmark and four dataset clusters at suite level, so exact tests are coarse and per-target intervals can be wide. The study uses one ImageNet-pretrained ResNet-50 recipe and four classification benchmarks. The SESOI and source-selected endpoint were fixed during revision rather than preregistered before the original study. Source validation is stochastic, and target curves existed in logs even though they do not enter the frozen selection rule. Native ERM and HyperDG have different BatchNorm/update routes. Most importantly, the implementation has an inactive local term, a scale-cancelling proxy, and unscaled inference, and the study does not include genuine manifold-valued training. These facts narrow the claim; they do not invalidate the negative result for the executed package.

Broader impact. This work uses standard image benchmarks and introduces no new human-subject data. Its main impact is methodological: reporting a carefully bounded negative result can prevent a scalar Euclidean implementation from being mistaken for evidence about hyperbolic learning as a whole. The same care is needed in the other direction; our results should not be cited to dismiss genuine hyperbolic representations.

8 Conclusion

We evaluated a specific curvature-inspired scalar feature-scaling package, not general hyperbolic representation learning. Across 384 completed runs, the full implementation showed no advantage over native ERM under the declared protocol, and its four dataset point estimates were lower. Within the shared update route, separate source-loader scales were equivalent to one shared learned scale for the source-selected suite macro within a revision-stage $\pm 1\,pp$ margin, but not for every dataset, target, or final-step endpoint. Together with the implementation audit, these results locate the negative finding precisely: it applies to weakly

identified training-time Euclidean rescaling with this ResNet-50 protocol and should not be generalized to manifold-valued hyperbolic models.

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